

Extremum problems

The largest volume, the least cost, the shortest path — modelling first, then one derivative.

LESSON 23–25

3 × 40 MIN

ALGEBRA 11, §1.8

P1 · 7.8

LESSON CLOCK

14'

26'

30'

26'

40'

14'

The five-step method	14 min
Geometric problems	26 min
Boxes and cans	30 min
Cost and time problems	26 min
Practice	40 min
Homework	14 min

LEARNING OBJECTIVES

- Express the quantity to be optimised as a function of one variable.
- Use a constraint to eliminate the second variable.
- Find the extremum and justify that it is a maximum or a minimum.
- Interpret the answer, checking it against the physical domain.

TEXTBOOK

National	pp. 111–126
Cambridge	Pure Mathematics 1, pp. 178–190
Workbook	P1 Exercise 7H

01

Explanation

Five steps

THE METHOD, EVERY TIME

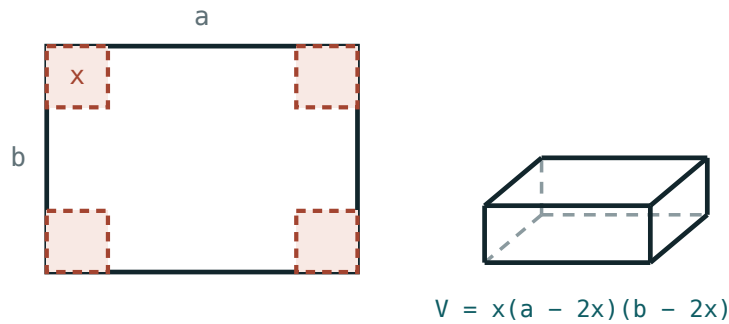
1 Name the variables and draw. **2** Write the quantity to optimise. **3** Use the constraint to reduce it to **one** variable, and state the feasible domain. **4** Differentiate, set to zero, solve. **5** Justify max or min, then answer in words with units.

STEP 3 IS WHERE MARKS ARE LOST

An expression in two variables cannot be differentiated. The constraint — the fixed perimeter, the fixed volume, the fixed length of wire — exists precisely to eliminate one of them.

The open box

Problem. A 24×24 cm square of card has equal squares of side x cut from its corners; the flaps fold up. Maximise the volume.



cut a square x from each corner

The card, the four corner squares removed, and the box that results.

$$V(x) = x(24 - 2x)^2, 0 < x < 12$$

$$V'(x) = (24 - 2x)^2 + x \cdot 2(24 - 2x)(-2) = (24 - 2x)(24 - 6x)$$

$V' = 0$ at $x = 12$ (rejected — outside the domain) and $x = 4$. Sign of V' : positive before 4, negative after, so $x = 4$ is a maximum.

$$V(4) = 4 \times 16^2 = 1024 \text{ cm}^3$$

The cheapest can

Problem. A closed cylindrical can holds 500 cm^3 . Minimise the surface area.

$$V = \pi r^2 h = 500 \Rightarrow h = \frac{500}{\pi r^2}$$

$$S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \frac{1000}{r}$$

$$S' = 4\pi r - \frac{1000}{r^2} = 0 \Rightarrow r^3 = \frac{250}{\pi} \Rightarrow r \approx 4.30 \text{ cm}$$

Then $h \approx 8.60 \text{ cm}$ — exactly twice the radius. $S'' = 4\pi + \frac{2000}{r^3} > 0$, so it is a minimum. $S \approx 349 \text{ cm}^2$.

THE RESULT IS WORTH REMEMBERING

For a closed cylinder of fixed volume, the cheapest shape always has $h = 2r$ — the height equals the diameter. Real cans are taller because the ends cost more per cm^2 than the wall.

Justifying, and the endpoints

A stationary point is not automatically the answer. Three ways to justify:

1. the sign of f' either side — always works;
2. f'' at the point — quick, but silent when it is zero;
3. compare with the **endpoint** values, when the domain is a closed interval.

ON A CLOSED INTERVAL, CHECK THE ENDS

The greatest value of a continuous function on $[a, b]$ occurs at a stationary point **or at an endpoint**. A problem asking for the maximum on $0 \leq x \leq 5$ needs $f(0)$ and $f(5)$ computed as well.

02 Worked examples

EXAMPLE 1 A farmer has 100 m of fence for a rectangular pen against a straight wall. Maximise the area.

① Sides $x, x, 100 - 2x$.
 $0 < x < 50$

② $A = x(100 - 2x) = 100x - 2x^2$

③ $A' = 100 - 4x = 0 \Rightarrow x = 25$

④ $A'' = -4 < 0$
 Maximum.

ANSWER

1250 m^2 , a 25 m by 50 m pen

EXAMPLE 2 Find the point on $y = x^2$ closest to $(0, 3)$.

① $D^2 = x^2 + (x^2 - 3)^2$

Minimise D^2 , not D .

② $= x^4 - 5x^2 + 9$

③ $(D^2)' = 4x^3 - 10x = 2x(2x^2 - 5) = 0$

④ $x = 0$ or $x = \pm\sqrt{2.5}$

$x = 0$ gives $D = 3$; the others $D \approx 1.66$

ANSWER

$(\pm\sqrt{2.5}, 2.5)$, at distance $\frac{\sqrt{11}}{2} \approx 1.66$

EXAMPLE 3 Find the greatest value of $f(x) = x^3 - 3x$ on $[0, 3]$.

① $f' = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$

Only $x = 1$ is in range.

② $f(1) = -2$

③ $f(0) = 0, f(3) = 18$

Endpoints.

④ The largest is at an endpoint.

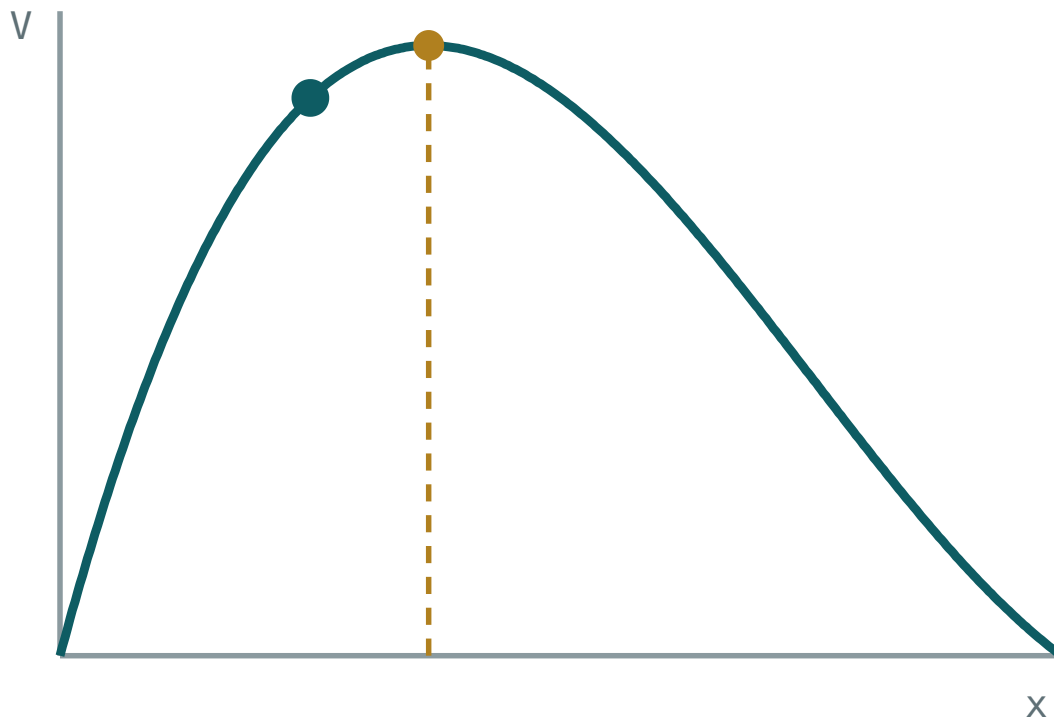
ANSWER

18, at $x = 3$

03 Interactive model

Slide x and watch the volume of the box rise and fall; mark the maximum.

The open box



x 2

$V(x)$ 384

greatest V 420.1

at $x = 2.94$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Extremum	Ekstremum	Экстремум
Greatest value	Eng katta qiymat	Наибольшее значение
Least value	Eng kichik qiymat	Наименьшее значение
Constraint	Cheklov	Ограничение
Objective function	Maqsad funksiyasi	Целевая функция
Feasible domain	Ruxsat etilgan soha	Допустимая область
Endpoint value	Chegaraviy qiymat	Значение на конце
Optimisation	Optimallashtirish	Оптимизация
Justification	Asoslash	Обоснование

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first thing to do in an extremum problem:

The constraint is used to:

On a closed interval the greatest value can occur:

The cheapest closed cylinder of fixed volume has:

$$h = r$$

$$h = 2r$$

$$h = 4r$$

$$r = 2h$$

To minimise a distance it is easier to minimise:

the distance

its square

its reciprocal

its logarithm

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 100 m of fence, rectangular field. Maximum area

ANSWER 625 m^2

2. Two numbers add to 20. Maximise the product

ANSWER 100

3. Two numbers add to 16. Minimise the sum of squares

ANSWER 128

4. Maximum of $y = 8x - x^2$

ANSWER 16 at $x = 4$

5. Minimum of $y = x^2 - 10x$

ANSWER -25 at $x = 5$

6. Domain for a corner cut x from a 20×20 card

ANSWER $0 < x < 10$

7. Justify a maximum using f''

ANSWER $f'' < 0$

Medium · the standard question

7 PROBLEMS

1. 80 m of fence against a wall. Maximum area

ANSWER 800 m^2

2. 30×30 card, corner squares cut. Maximum volume

ANSWER 2000 cm^3 at $x = 5$

3. Closed cylinder, volume 1000 cm^3 . Minimum surface area

ANSWER $\approx 554 \text{ cm}^2$

4. Open cylinder (no lid), volume 500. Optimal radius

ANSWER $\approx 5.42 \text{ cm}$

5. Greatest value of $y = x^3 - 3x^2$ on $[0, 4]$

ANSWER 16 at $x = 4$

6. Least value of $y = x + \frac{9}{x}$ for $x > 0$

ANSWER 6 at $x = 3$

7. Rectangle of perimeter 40 with maximum area

ANSWER 100, a 10-square

Hard · stretch and reason

7 PROBLEMS

1. A rectangle inscribed in a circle of radius 5. Maximise the area

ANSWER 50, a square of side $5\sqrt{2}$

2. Point on $y = x^2$ closest to $(0, 3)$

ANSWER $(\pm\sqrt{2.5}, 2.5)$

3. A cone of slant height 6. Maximise its volume

ANSWER $r = 2\sqrt{6}$, $V = 16\pi\sqrt{3}$

4. Wire of 60 cm cut into a square and a circle. Minimise the total area

ANSWER Square ≈ 33.6 cm of wire

5. Cheapest open-topped box of square base and volume 32 cm^3

ANSWER base 4 cm, height 2 cm

6. A window is a rectangle topped by a semicircle, perimeter 10 m. Maximise the area

ANSWER $r = \frac{10}{\pi + 4} \approx 1.40 \text{ m}$

7. A man rows at 4 km/h and walks at 5 km/h from a point 3 km offshore to a point 8 km along. Minimise the time

ANSWER Land 4 km along the shore

07 Homework

Homework — 6 tasks

Every answer needs the justification and the units.

- 120 m of fence, rectangular pen against a wall. Find the greatest area.
- A 40×40 cm card: corner squares are cut and the flaps folded. Find the greatest volume.
- A closed cylindrical can holds 750 cm^3 . Find the radius that minimises the surface area, and check $h = 2r$.

4. Find the greatest and least values of $y = x^3 - 6x^2 + 9x$ on $[0, 4]$.
5. Find the point on $y = \sqrt{x}$ closest to $(4, 0)$.
6. Two numbers have a sum of 30. Maximise the product of one and the square of the other.